

LECTURE NOTES MATH255

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1. LECTURE 11

In this class we discuss field extensions, and we use the language of field extensions to show that every polynomial has a zero.

1.1. Field extensions. The definition of a field extension of a field F , is any field E that contains F :

Definition 1.1. Let F be a field. A field extension of F is a field E such that F is a subfield of E , $F \leq E$.

Example 1.1. Notice that we have $\mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$, so $\mathbb{R}$ is a field extension of $\mathbb{Q}$, $\mathbb{C}$ is a field extension of $\mathbb{R}$, and $\mathbb{C}$ is a field extension of $\mathbb{Q}$.

Example 1.2. The fields:

$$(1.1) \quad \mathbb{Q}(\sqrt{7}) = \{x + y\sqrt{7} \mid x, y \in \mathbb{Q}\},$$

$$(1.2) \quad \mathbb{Q}(i) = \{x + yi \mid x, y \in \mathbb{Q}\},$$

are field extensions of $\mathbb{Q}$ since we can set $y = 0$ and obtain $\mathbb{Q}$ as a subfield of both of these fields. In fact, if $q \in \mathbb{Q}$ is arbitrary then:

$$(1.3) \quad \mathbb{Q}(\sqrt{q}) = \{x + y\sqrt{q} \mid x, y \in \mathbb{Q}\}$$

is always a field extension of $\mathbb{Q}$ (though, we might have $\mathbb{Q}(\sqrt{q}) = \mathbb{Q}$, this happens when q is a square: $q = a^2$ with $a \in \mathbb{Q}$).

Theorem 1.1. Let $f(x) \in F[x]$ be a non-constant polynomial with coefficients in F . There exists a field extension $F \leq E$ such that there is some $\alpha \in E$ that is a zero for $f(x)$, $f(\alpha) = 0$. That is, any polynomial $f(x) \in F[x]$ has a zero in some field extension.

Proof. We want to begin by reducing the theorem to the case when $f(x)$ is irreducible. Let $f(x) \in F[x]$ be arbitrary, and let $f(x) = f_1(x) \dots f_n(x)$ be the decomposition of $f(x)$ into irreducible polynomials $f_j(x) \in F[x]$ over F . Notice that if $F \leq E$ and if $f_1(\alpha) = 0$ for some $\alpha \in E$ then:

$$(1.4) \quad f(\alpha) = f_1(\alpha)f_2(\alpha)\dots f_n(\alpha) = 0 \cdot f_2(\alpha)\dots f_n(\alpha) = 0$$

so $f(x)$ also has a zero at $\alpha \in E$. That is, after replacing $f(x)$ by $f_1(x)$ we may assume without loss of generality that $f(x) \in F[x]$ is irreducible over F .

Since $f(x) \in F[x]$ is irreducible we know that $\langle f(x) \rangle$ is a maximal ideal, and since $\langle f(x) \rangle$ is a maximal ideal we know that $E = F[x]/\langle f(x) \rangle$ is a field. We define a map $\psi : F \rightarrow E$ by $\psi(a) = a + \langle f(x) \rangle$ for $a \in F$ (where we understand $a \in F$ as the constant polynomial $g(x) = a$). Note that this map is a homomorphism of fields:

$$(1.5) \quad \psi(a + b) = a + b + \langle f(x) \rangle = (a + \langle f(x) \rangle) + (b + \langle f(x) \rangle) = \psi(a) + \psi(b),$$

$$(1.6) \quad \psi(ab) = ab + \langle f(x) \rangle = (a + \langle f(x) \rangle)(a + \langle f(x) \rangle) = \psi(a)\psi(b),$$

$$(1.7) \quad \psi(1) = 1 + \langle f(x) \rangle$$

and the unity in E is $1 + \langle f(x) \rangle$. Moreover, $\psi : F \rightarrow E$ is an injective homomorphism of fields. We can in fact prove this in much more generality: any homomorphism between fields is injective. To see this, let $\varphi : K \rightarrow L$ be any homomorphism between fields K and L . To show that φ is injective it suffices to show that $\ker \varphi = \{0_K\}$. To see this, note that if $\varphi(a) = 0$ and $a \neq 0_K$ then:

$$(1.8) \quad 1_L = \varphi(1_K) = \varphi(a \cdot a^{-1}) = \varphi(a)\varphi(a^{-1}) = 0_L \cdot \varphi(a^{-1}) = 0_L$$

but in a field we have $1_L \neq 0_L$ so this is a contradiction. We conclude that if $\varphi(a) = 0_L$ then $a = 0_K$, so $\ker \varphi = \{0_K\}$ and therefore φ is injective. It follows, in particular, that the map $\psi : F \rightarrow E$ is an injective field homomorphism. After identifying F with its image in E (note that $\psi : F \rightarrow \text{Im}(\psi)$ is an isomorphism since ψ is injective), $\text{Im}(\psi) \leq E$ we may view F as a subfield of E . In this sense, E is an extension field of F . It remains to check that there is some $\alpha \in E$ such that $f(\alpha) = 0$. To see this, let $\alpha = x + \langle f(x) \rangle$ and write $f(x) = a_0 + \dots + a_n x^n$ where $a_0, \dots, a_n \in F$. We have:

$$(1.9) \quad f(\alpha) = f(x + \langle f(x) \rangle) = a_0 + a_1(x + \langle f(x) \rangle) + \dots + a_n(x + \langle f(x) \rangle)^n =$$

$$(1.10) \quad a_0 + a_1 x + \dots + a_n x^n + \langle f(x) \rangle = f(x) + \langle f(x) \rangle = 0 + \langle f(x) \rangle$$

where the last equality use that $f(x) \in \langle f(x) \rangle$ since $f(x) = f(x) \cdot 1$ so $f(x) + \langle f(x) \rangle = 0 + \langle f(x) \rangle$. ■

Example 1.3. To make the construction of an extension field $F \leq E$ where a polynomial $f(x) \in F$ has a zero a little bit more explicit we will look at the example of $f(x) = x^2 + 1 \in \mathbb{R}[x]$. Note that $f(x)$ is irreducible since its quadratic with zeros $\pm i \notin \mathbb{R}$. It follows that the extension field produced in the theorem should be:

$$(1.11) \quad E = \mathbb{R}[x]/\langle x^2 + 1 \rangle.$$

We claim that any $p(x) + \langle x^2 + 1 \rangle \in E$ can be represented by a polynomial $p(x) = ax + b$ of degree at most 1. Indeed, there are multiple ways to see this, but one way is to note that if $p(x)$ is any polynomial then we can write using the division algorithm: $p(x) = (x^2 + 1)q(x) + r(x)$ where $r(x) = ax + b$ should have degree at most 1 (since $\deg(r(x)) < \deg(x^2 + 1) = 2$). However, $(x^2 + 1)q(x) \in \langle x^2 + 1 \rangle$ so:

$$\begin{aligned} p(x) + \langle x^2 + 1 \rangle &= (x^2 + 1)q(x) + r(x) + \langle x^2 + 1 \rangle = r(x) + \langle x^2 + 1 \rangle = \\ &= ax + b + \langle x^2 + 1 \rangle. \end{aligned}$$

Notice that $(x + \langle x^2 + 1 \rangle)^2 = x^2 + \langle x^2 + 1 \rangle = x^2 + 1 - 1 + \langle x^2 + 1 \rangle = -1 + \langle x^2 + 1 \rangle$, so the square of $x + \langle x^2 + 1 \rangle$ is $-1 + \langle x^2 + 1 \rangle$. Motivated by this we can define a map $\psi : E \rightarrow \mathbb{C}$ by $\psi(ax + b + \langle x^2 + 1 \rangle) = ai + b \in \mathbb{C}$, and because $(x + \langle x^2 + 1 \rangle)^2 = -1 + \langle x^2 + 1 \rangle$ one can prove directly that ψ is an isomorphism of fields. Another way to see that

ψ is an isomorphism of fields is to note that we could have defined $\psi : E \rightarrow \mathbb{C}$ by $\psi(p(x) + \langle x^2 + 1 \rangle) = p(i)$ which is well-defined since if $p(x) + \langle x^2 + 1 \rangle = \tilde{p}(x) + \langle x^2 + 1 \rangle$ then $\tilde{p}(x) = (x^2 + 1)q(x) + p(x)$, so $\tilde{p}(i) = (i^2 + 1)q(i) + p(i) = p(i)$. Moreover, we see that ψ is a field homomorphism:

$$\psi(p(x) + q(x) + \langle x^2 + 1 \rangle) = p(i) + q(i) = \psi(p(x) + \langle x^2 + 1 \rangle) + \psi(q(x) + \langle x^2 + 1 \rangle),$$

and

$$\begin{aligned} \psi((p(x) + \langle x^2 + 1 \rangle)(q(x) + \langle x^2 + 1 \rangle)) &= \psi(p(x)q(x) + \langle x^2 + 1 \rangle) = q(i)p(i) = \\ &= \psi(p(x) + \langle x^2 + 1 \rangle)\psi(q(x) + \langle x^2 + 1 \rangle), \end{aligned}$$

and $\psi(1 + \langle x^2 + 1 \rangle) = 1$. It follows that ψ is injective, since it's a homomorphism between fields, but we also note that it's surjective since for any $a + bi \in \mathbb{C}$ we have $\psi(a + bx + \langle x^2 + 1 \rangle) = a + ib$. That is, we have that $\mathbb{R}[x]/\langle x^2 + 1 \rangle \cong \mathbb{C}$.

Definition 1.2. Let E be an extension field of F , $F \leq E$. An element $\alpha \in E$ is algebraic over F if there is a non-constant polynomial $f(x) \in F[x]$ such that $f(\alpha) = 0$. If $\alpha \in E$ is not algebraic over F then we say that α is transcendental over F .

Example 1.4. The numbers $\pi, e \in \mathbb{C}$ are not examples of algebraic numbers over $\mathbb{Q}$, they are transcendental over $\mathbb{Q}$ (we will not prove this in this course).

Example 1.5. The number $\sqrt{7} \in \mathbb{C}$ is algebraic over $\mathbb{Q}$ since it's the zero of $x^2 - 7 \in \mathbb{Q}[x]$.

Definition 1.3. A complex number $\alpha \in \mathbb{C}$ is algebraic if it is algebraic over $\mathbb{Q}$. A complex number $\alpha \in \mathbb{C}$ is transcendental if it is transcendental over $\mathbb{Q}$.

Theorem 1.2. Let $F \leq E$ be an extension field of F . If $\alpha \in E$ is algebraic over F then there exists an irreducible polynomial $f(x) \in F[x]$ such that $f(\alpha) = 0$. Moreover, $f(x)$ is unique up to multiplication by $a \in F \setminus \{0\}$. If $g(x) \in F[x]$ satisfies $g(\alpha) = 0$ then $f(x)$ divides $g(x)$. We can find $f(x)$ as any non-constant polynomial of minimal degree such that $f(\alpha) = 0$.

Proof. Let α be as in the theorem and let $f(x) \in F[x]$ be any non-constant polynomial of minimal degree such that $f(\alpha) = 0$ (note that there are non-constant polynomials $f(x)$ such that $f(\alpha) = 0$ exists since α is algebraic by assumption). We begin by showing that $f(x)$ is irreducible. Suppose that $f(x)$ was not irreducible, then we could write $f(x) = g(x)h(x)$ where $\deg(g(x)), \deg(h(x)) < \deg(f(x))$, and both $g(x), h(x)$ non-zero since $f(x)$ is not zero. However, we have $0 = f(\alpha) = g(\alpha)h(\alpha)$ so, since E is a field and therefore an integral domain, we conclude that $g(\alpha) = 0$ or $h(\alpha) = 0$. But this would contradict that $f(x)$ is chosen with a minimal degree such that $f(\alpha) = 0$, since both $g(x)$ and $h(x)$ have a lower degree than $f(x)$. With this contradiction, we conclude that $f(x)$ is irreducible.

Let $g(x)$ be any other polynomial such that $g(\alpha) = 0$. By the division algorithm we can write $g(x) = f(x)q(x) + r(x)$ where $\deg(r(x)) < \deg(f(x))$. By evaluating at α we obtain $0 = g(\alpha) = f(\alpha)q(\alpha) + r(\alpha) = r(\alpha)$ so $r(\alpha) = 0$. This implies that $r(x) = 0$, since otherwise $r(x)$ would be a non-constant polynomial with a zero at α which would contradict the minimal degree of $f(x)$. We conclude that $g(x) = q(x)f(x)$, so $f(x)$ divides $g(x)$. We can also use this to prove that $f(x)$ is the unique (up to

scaling) irreducible polynomial for which α is a zero: if $g(x) \in F[x]$ is irreducible and $g(\alpha) = 0$, then $g(x) = f(x)q(x)$ irreducibility of $g(x)$ then implies that $q(x)$ must be constant so $g(x) = af(x)$ where $a \in F \setminus \{0\}$. ■

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